THE PHYSICAL AI CURRICULUM BLUEPRINT & ROADMAP

Systematic Deconstruction: From Causal Foundations to Pipeline Organs & Real-World Assurance

PART 1: THE FOUNDATIONAL TRIAD (THE REALM OF PHYSICS & LATENCY)

CHAPTER 1
Physical Causality
Causal boundary, 3 criteria,
irreversible mutation ($W_t \rightarrow W_{t+1}$)

CHAPTER 2
Time & Latency Metrology
7-stage latency ledger, tail P99,
freshness wall & stopping distance

CHAPTER 3
The Agent Workflow
Great Tug-of-War, 9-station lifecycle,
3 cadences & multi-rate contracts

PART 2: THE 7 CANONICAL PIPELINE ORGANS (BUILDING THE WORKFLOW)

CHAPTER 4
Perception
MIPI DMA stream
ViT / DINOv2 tokens
3D SE(3) affordances
[Stations 2 & 3]

CHAPTER 5
World Models
Latent JEPAs
Coordinate trees
Uncertainty bounds
[Station 4]

CHAPTER 6
Semantic Intent
1 Hz VLMs
Open-world goals
Expiring leases (TTL)
[Stations 1 & 5]

CHAPTER 7
Action Chunking
Diffusion / ACT
Delay amortization
C^2 jerk continuous
[Station 6]

CHAPTER 8
Real-Time Reflex
1 kHz CBF safety QP
Proposal-permission
IEC dynamic halts
[Station 7]

PART 3: INTEGRATION, GOVERNANCE & DEPLOYMENT (SYSTEM QUALIFICATION)

CHAPTER 9
Workload Placement
MPU vs MCU vs NPU
UMA memory contention & IPC

CHAPTER 10
Human Governance
Bumpless manual takeover
Telemetry logs & intervention tags

CHAPTER 11
Assurance & Release
STPA hazard mitigation
Claim-Argument-Evidence case

CHAPTER 12
Capstone Deployment
Full dual-brain release
Arduino UNO Q bench qualification